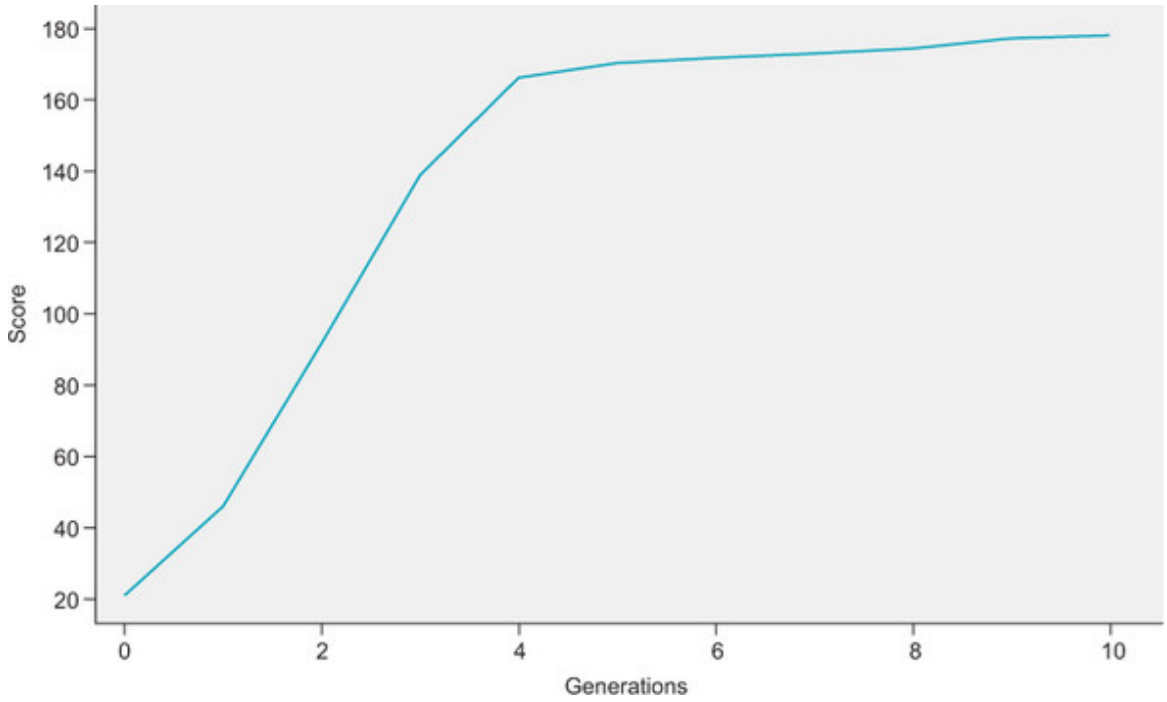


6.4. Pros and cons of evolutionary algorithms

The algorithm we implemented in this chapter is a bit different from the previous approaches we've used in this book. There are circumstances where an evolutionary approach works better, such as with problems that would benefit more from exploration; other circumstances make it impractical, such as problems where it is expensive to gather data. In this section we'll discuss the advantages and disadvantages of evolutionary algorithms and where you might benefit from using them over gradient descent.

6.4.1. Evolutionary algorithms explore more

One advantage of gradient-free approaches is that they tend to explore more than their gradient-based counterparts. Both DQN and policy gradients followed a similar strategy: collect experiences and nudge the agent to take actions that led to greater rewards. As we discussed, this tends to cause agents to abandon exploring new states if they prefer an action already. We addressed this with DQN by incorporating an epsilon-greedy strategy, meaning there's a small chance the agent will take a random action even if it has a preferred action. With the stochastic policy gradient we relied on